

Impulse & Momentum

①

Impulse of force $\vec{F}$ is given as

$$J = \int_{t_1}^{t_2} \vec{F} dt$$

Recall $\vec{F} = \frac{d\vec{p}}{dt}$ } from Newton's second law
 $\vec{F}$ is the rate of change of momentum ($\vec{p}$)

$$\therefore J = \Delta \vec{p}$$

Impulse of a force $\vec{F}$ is the change in the momentum of the body.
↳ Impulse-momentum Theorem.

- 1) A cricket ball has an initial velocity $v_o = -38 \text{ m/s}$ as it approaches a bat, assuming the direction of approach is -ve. Once the ball leaves the bat $v_f = +58 \text{ m/s}$. If $m_{\text{ball}} = 0.14 \text{ kg}$, determine:
- the impulse applied by bat on the ball
 - if time for the impact is $1.6 \times 10^{-3} \text{ s}$, find the force exerted on the ball.

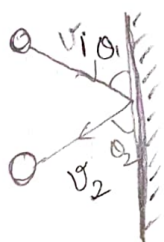
Sol

$$\begin{aligned} \text{a) } J &= \Delta \vec{p} = m v_f - m v_i \\ &= m (v_f - v_i) \\ &= 0.14 (58 - (-38)) = 13.4 \text{ kg m/s} \end{aligned}$$

b) $\vec{J} = \int_{t_1}^{t_2} \vec{F} \cdot dt = F_{avg} \cdot \Delta t$

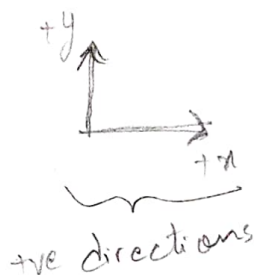
$$F_{avg} = \frac{\vec{J}}{\Delta t} = \frac{13.4 \text{ kg m/s}}{1.6 \times 10^{-3} \text{ s}} = \underline{\underline{8400 \text{ N}}}$$

2)

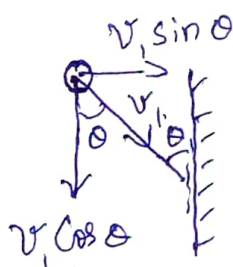


A ball of mass 0.5 kg hits the wall with a velocity of 20 m/s & at angle $\theta_1 = 60^\circ$. The impact lasts for 0.02 s (20 milliseconds).

The ball rebounds with a speed of 15 m/s at angle $\theta_2 = 45^\circ$ to the wall. Determine the change in momentum & the Average force on the ball.



Sol

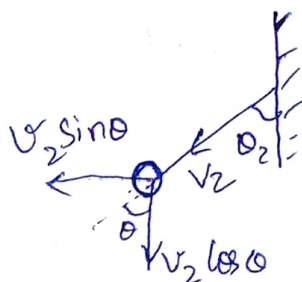


} Before impact.

$$\vec{v}_1 = v_1 \sin \theta_1 \hat{i} - v_1 \cos \theta_1 \hat{j}$$

$$\vec{v}_1 = 20 \cdot \sin 60^\circ \hat{i} - 20 \cos 60^\circ \hat{j}$$

$$\boxed{\vec{v}_1 = (17.32 \hat{i} - 10 \hat{j}) \text{ m/s}}$$



} After impact.

$$\vec{v}_2 = -v_2 \sin \theta_2 \hat{i} - v_2 \cos \theta_2 \hat{j}$$

$$\vec{v}_2 = -15 \sin 45^\circ \hat{i} - 15 \cos 45^\circ \hat{j}$$

$$\vec{v}_2 = (-10.61 \hat{i} - 10.61 \hat{j}) \text{ m/s}$$

$\therefore$ Change in momentum $\Delta \vec{p} = m (\vec{v}_2 - \vec{v}_1)$

$$\therefore \Delta \vec{p} = m [(-10.61 \hat{i} - 10.61 \hat{j}) - (17.32 \hat{i} - 10 \hat{j})]$$

$$= 0.5 [(-10.61 - 17.32) \hat{i} + (-10.61 + 10) \hat{j}]$$

$$= 0.5 [-27.93 \hat{i} - 0.61 \hat{j}]$$

$$|\Delta \vec{p}| = 0.5 \sqrt{(27.93)^2 + (0.61)^2} = \underline{\underline{13.97 \text{ kg m/s}}}$$

Impact for 0.02 s

$$J = \vec{F} \cdot \Delta t \Rightarrow \vec{F} = \frac{J}{\Delta t} = \frac{\Delta \vec{p}}{\Delta t} = \frac{13.97}{0.02} = \underline{\underline{698.5 \text{ N}}}$$

Conservation of linear momentum

$\rightarrow$ In case of two bodies ① & ② such that they interact as below:

$$\frac{d\vec{p}_1}{dt} = \vec{F}_{\text{net}} \text{ on body 1}$$

$\vec{F}_{\text{net}}$ on body 1 is $\vec{F}_{12} + \vec{F}_1^{\text{external}}$ (4)

force exerted on 1 by 2

any external force acting on 1

$$\therefore \frac{d\vec{p}_1}{dt} = \vec{F}_{12} + \vec{F}_1^{\text{external}}$$

Similarly $\frac{d\vec{p}_2}{dt} = \vec{F}_{21} + \vec{F}_2^{\text{external}}$

force exerted on 2 by 1

$\therefore$ For the 2 particle system net momentum change is

$$\frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} = \frac{d(\vec{p}_1 + \vec{p}_2)}{dt}$$

$$\therefore \frac{d(\vec{p}_1 + \vec{p}_2)}{dt} = \vec{F}_{12} + \vec{F}_{21} + (\vec{F}_1 + \vec{F}_2)^{\text{external}}$$

If the system of 2 particles has no external forces the $(\vec{F}_1 + \vec{F}_2)^{\text{external}} = 0$.

From Newton's 3rd law

$$\vec{F}_{12} = -\vec{F}_{21} \quad \left. \begin{array}{l} \text{Action on body 1 is equal} \\ \text{to the reaction on body 2 but} \\ \text{opposite in direction.} \end{array} \right\}$$

$$\therefore \vec{F}_{12} + \vec{F}_{21} = 0$$

(5)

$$\therefore \frac{d(\vec{p}_1 + \vec{p}_2)}{dt} = \underbrace{\vec{F}_{12} + \vec{F}_{21}}_0 + \underbrace{(\vec{F}_1 + \vec{F}_2)}_0^{\text{external}}$$

(from Newton's 3rd law) (from no external forces acting on system)

$$\Rightarrow \frac{d(\vec{p}_1 + \vec{p}_2)}{dt} = 0$$

$$\Rightarrow \vec{p}_1 + \vec{p}_2 = \text{Constant}$$

This suggests that the net momentum of the system is conserved for a 2 particle system with no external forces.

→ Now Consider N masses in the system & that there are no external forces:



for such a system.

$$\frac{d\vec{p}_1}{dt} = \vec{F}_{12} + \vec{F}_{13} + \dots \dots \dots \vec{F}_{1N}$$

$$\frac{d\vec{p}_2}{dt} = \vec{F}_{21} + \vec{F}_{23} + \dots \dots \dots \vec{F}_{2N}$$

⋮

$$\frac{d\vec{p}_N}{dt} = \vec{F}_{N1} + \vec{F}_{N2} + \dots \dots \dots \vec{F}_{N,N-1}$$

Rate of change of momentum for each particle is the sum of all forces on that particle due to other particles.

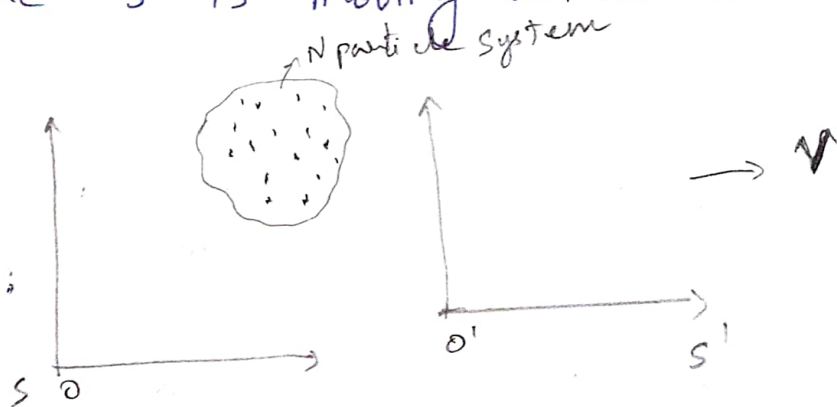
$$\therefore \frac{d(\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_N)}{dt} = \underbrace{(\vec{F}_{12} + \vec{F}_{21})}_0 + \underbrace{(\vec{F}_{13} + \vec{F}_{31})}_0 + \dots + \underbrace{(\vec{F}_{N-1,N} + \vec{F}_{N,N-1})}_0 \quad (6)$$

$$\therefore \frac{d(\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_N)}{dt} = \frac{d(\sum_N \vec{p}_i)}{dt} = 0$$

$$\Rightarrow \sum_N \vec{p}_i = \text{Constant}$$

This is the conservation of linear momentum for N particles in a system without any external forces.

→ Now consider 2 frames of reference as earlier S & S' such that S is stationary while S' is moving with constant velocity " v "



$$\sum_N \vec{p}_i = \sum_N m_i \vec{v}_i = \text{Constant} \quad \left\{ \begin{array}{l} \text{no external} \\ \text{forces} \end{array} \right.$$

momentum conservation for " S " frame of reference is applicable for N particles.

Recall

$$\vec{v}' = \vec{v} - \vec{V}$$

$$\left. \begin{array}{l} \text{velocity of} \\ \text{Particle in } s' \\ \text{frame} \end{array} \right\} = \left. \begin{array}{l} \text{velocity of} \\ \text{Particle in} \\ s \text{ frame} \end{array} \right\} - \text{velocity of } s' \text{ frame.}$$

Now the net momentum of N particle system in s' frame of reference can be given as

$$\begin{aligned} \sum_N \vec{p}_i' &= \sum_N m \vec{v}_i' \\ &= \sum_N m (\vec{v}_i - \vec{V}) \end{aligned}$$

$$\sum_N \vec{p}_i' = \sum_N m \vec{v}_i - \sum_N m \vec{V}$$

$$\sum_N \vec{p}_i' = \sum_N \vec{p}_i - \sum_N m \vec{V}$$

→ derivate with respect to time.

$$\frac{d \sum_N \vec{p}_i'}{dt} = \frac{d \sum_N \vec{p}_i}{dt} - \frac{\sum_N d(m \vec{V})}{dt}$$

from conservation
of momentum in s
frame of reference

m & V are
constant
⇒ this derivative
is 0

$$\therefore \underbrace{\frac{d \sum_N \vec{p}_i}{dt}} = \underline{0}$$

$$\Rightarrow \sum_N \vec{p}_i = \text{Constant}$$

$\therefore$ The conservation of momentum holds true for an inertial frame of reference as long as no external forces are acting on the system of particles.